

E - LRUD Moving

分值: 450 分

Problem Statement

You are given positive integers N, A, B . It is guaranteed that both A and B are between 1 and N , inclusive.

给定正整数 N, A, B 。保证 A 和 B 均介于 1 和 N 之间（含两端）。

There is an $N \times N$ grid. The cell at the i -th row from the top and j -th column from the left is denoted as cell (i, j) . Initially, a piece is placed at cell $(1, 1)$.

有一个 $N \times N$ 的网格。从上往下第 i 行、从左往右第 j 列的单元格记为 (i, j) 。初始时，一枚棋子被放置在单元格 $(1, 1)$ 。

By repeating $N^2 - 2$ moves, each of which moves the piece to an adjacent cell (up, down, left, or right), you want to move the piece to cell (N, N) while visiting every cell except cell (A, B) . You must not visit the same cell more than once (you must not visit cells $(1, 1)$ and (N, N) in the middle, either).

你需要通过重复 $N^2 - 2$ 次移动，每次将棋子移至相邻单元格（上、下、左、右之一），将棋子移动到单元格 (N, N) ，同时访问除 (A, B) 外的所有单元格。你不能重复访问同一个单元格（移动过程中也不能访问 $(1, 1)$ 和 (N, N) ）。

Determine whether such a sequence of moves is possible, and if so, output one such sequence.

判断是否存在这样的移动序列。若存在，输出其中一种合法序列。

You are given T test cases; solve each of them.

给定 T 组测试用例，你需要分别处理每组数据。

Constraints

- $1 \leq T \leq 5000$
- $2 \leq N \leq 10^3$

- $1 \leq A, B \leq N$
 - $(A, B) \neq (1, 1), (N, N)$
 - The sum of N^2 over all test cases is at most 10^6 .
 - All input values are integers.
 - $1 \leq T \leq 5000$
 - $2 \leq N \leq 10^3$
 - $1 \leq A, B \leq N$
 - $(A, B) \neq (1, 1), (N, N)$
 - 所有测试用例的 N^2 之和不超过 10^6 。
 - 所有输入值均为整数。
-

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

```
T
case1
case2
⋮
caseT
```

Each test case is given in the following format:

每组测试用例按照以下格式给出：

```
N A B
```

Output

Output the answers for the test cases in order, separated by newlines.

按顺序输出每个测试用例的答案，答案之间用换行分隔。

For each test case, if it is impossible to perform a sequence of moves satisfying the conditions, output No.

对于每组测试用例，如果不存在满足条件的移动序列，输出 No。

If it is possible, output in the following format:

如果存在，按照以下格式输出：

Yes

$S_1 S_2 \dots S_{N^2-2}$

Yes

$S_1 S_2 \dots S_{N^2-2}$

Here, S_k is defined as follows, where the coordinates of the piece before the k -th move is cell (i, j) .

其中 S_k 的定义如下：第 k 次移动前棋子所在的坐标为单元格 (i, j) 。

- $S_k = L$ if the piece moves from cell (i, j) to cell $(i, j - 1)$
- $S_k = R$ if the piece moves from cell (i, j) to cell $(i, j + 1)$
- $S_k = U$ if the piece moves from cell (i, j) to cell $(i - 1, j)$
- $S_k = D$ if the piece moves from cell (i, j) to cell $(i + 1, j)$
- 若棋子从单元格 (i, j) 移动到单元格 $(i, j - 1)$ ，则 $S_k =$ 为 L
- 若棋子从单元格 (i, j) 移动到单元格 $(i, j + 1)$ ，则 $S_k =$ 为 R
- 若棋子从单元格 (i, j) 移动到单元格 $(i - 1, j)$ ，则 $S_k =$ 为 U
- 若棋子从单元格 (i, j) 移动到单元格 $(i + 1, j)$ ，则 $S_k =$ 为 D

If multiple sequences of moves satisfy the conditions, any of them will be accepted.

如果存在多个满足条件的移动序列，你可以输出任意一个。

Sample Input 1

```
3
2 1 2
3 2 2
4 3 2
```

Sample Output 1

```
Yes
DR
No
Yes
RRRDLLLDRRURD
```

Consider the first test case.

考虑第一组测试用例。

Starting with the piece at cell (1, 1), move it twice as follows:

初始时棋子位于单元格 (1, 1)，按如下方式移动两次：

- Move the piece downward. The piece moves to cell (2, 1).
- Move the piece rightward. The piece moves to cell (2, 2).
- 将棋子向下移动，棋子移动到单元格 (2, 1)。
- 将棋子向右移动，棋子移动到单元格 (2, 2)。

This sequence of moves satisfies the conditions.

该移动序列满足条件。